

PREDICTIVE ABS THROUGH PREDICTIVE WHEEL LOCK AND PRESSURE MODULATION

S2-25_SSZG628T: Dissertation

By

GOPINATH MADHESWARAN

2024MT12082

Dissertation work carried out at

AUMOVIO INDIA PVT LTD (Former CONTINENTAL AUTOMOTIVE), Bangalore

Submitted in partial fulfilment of **M.Tech degree programme**

Under the Supervision of **Vaidyanathan AG**

AUMOVIO INDIA PVT LTD, Bangalore



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE PILANI (RAJASTHAN)

(March 2026)

ABSTRACT

Anti-lock Braking Systems (ABS) play an essential role in improving vehicle stability and braking safety. Conventional ABS methods operate based on reactive control, where brake pressure modulation is initiated after the onset of excessive wheel slip. Although widely adopted, such reactive approaches may lead to oscillatory braking behaviour and delayed response during rapid braking conditions.

The present work focuses on the development of a **predictive anti-lock braking strategy using multi-feature signal analysis techniques**. The proposed approach utilizes wheel speed data sampled at **1000 Hz** to compute slip ratio and statistical indicators that provide early signs of impending wheel lock.

Three primary signal features are evaluated in this work:

- Approximate Entropy
- Statistical Kurtosis
- Dominant Frequency Estimation

These features are combined to generate a **weighted lockup probability**, which enables early prediction of unstable wheel behaviour before traditional ABS activation.

The algorithm has been implemented using Python-based signal processing tools and evaluated using a real-world braking dataset consisting of **15,000 samples across four wheels**. Simulation and validation results demonstrate earlier lockup detection and improved braking efficiency when compared with conventional ABS logic.

The findings indicate that predictive feature-based methods can improve braking smoothness, reduce stopping distance, and enhance slip regulation under dynamic driving conditions.

Signature of the Student

Name: Gopinath Madheswaran

Date: 26-03-2026

Place: Bangalore

Signature of the Supervisor

Name:

Date:

Place:

1. MODULES IN PREDICTIVE ABS SYSTEM

The predictive ABS architecture consists of several interconnected functional modules that enable signal acquisition, feature computation, prediction logic, and braking control evaluation.

The major modules in the system are:

- a) Wheel Speed Data Acquisition
- b) Signal Conditioning and Buffer Management
- c) Slip Ratio Computation
- d) Feature Extraction Module
- e) Lockup Probability Estimation
- f) Predictive Warning Generation
- g) Performance Evaluation Module

Each module is described below.

a) Wheel Speed Data Acquisition

Wheel speed measurements are obtained from a dataset representing braking behaviour under various surface conditions. The dataset consists of rotational speed values for four wheels: Front Left (FL), Front Right (FR), Rear Left (RL), and Rear Right (RR).

The signals are sampled at a rate of **1000 Hz**, allowing fine temporal resolution during braking events. The dataset also includes brake pressure information and reference vehicle speed values used for slip estimation.

b) Signal Conditioning and Buffer Management

Incoming wheel speed data is processed through a fixed-size signal buffer that stores recent samples required for feature computation. A sliding window of **20 samples** is used to analyse signal behaviour over short time intervals.

Basic signal normalization and filtering steps are applied to reduce noise and improve stability of the computed features.

This buffering mechanism ensures consistent feature evaluation during continuous braking operation.

c) Slip Ratio Computation

Slip ratio is calculated using vehicle speed and wheel speed signals. This parameter represents the difference between vehicle motion and wheel rotational speed.

Slip ratio values are continuously monitored to determine whether the wheel operates within the optimal friction region.

Typical optimal slip values lie between **10% and 20%**, which corresponds to maximum tire-road friction conditions.

Wheel Slip Definition

The normalized longitudinal slip ratio is defined as:

$$\lambda = \frac{v_{vehicle} - v_{wheel}}{v_{vehicle}}$$

where $v_{vehicle}$ is the vehicle reference velocity (typically from speedometer or GPS) and v_{wheel} is the wheel rotational velocity converted to linear units. This dimensionless metric captures the relative motion between the tire and road:

- $\lambda = 0$: Free rolling (no braking)
- $\lambda \in [0.10, 0.20]$: Optimal operating region for most tire-road interfaces
- $\lambda = 1.0$: Fully locked wheel (zero rotation)

d) Feature Extraction Module

The feature extraction stage computes multiple statistical indicators that characterize the behaviour of wheel speed signals.

The following features are evaluated:

Approximate Entropy

Approximate entropy measures the irregularity present in wheel speed signals. As a wheel approaches lock conditions, the signal behaviour changes and entropy values exhibit noticeable variations.

This feature is computed using sliding window samples and tolerance-based pattern comparison methods.

Approximate Entropy (ApEn)

Approximate entropy is a regularity statistic that quantifies the unpredictability of fluctuations in a time series. For a discrete signal $\{u_i\}_{i=1}^N$, ApEn is defined as:

$$\text{ApEn}(m, r, N) = \phi(m) - \phi(m + 1)$$

where:

$$\phi(m) = \frac{1}{N - m + 1} \sum_{i=1}^{N-m+1} \log C_i^m(r)$$

and $C_i^m(r)$ is the relative frequency of template matches. Intuitively, a complex or irregular signal has high ApEn, while a regular or predictable signal has low ApEn.

Statistical Kurtosis

Kurtosis is used to quantify the presence of sharp variations in signal distribution.

During stable braking conditions, wheel speed variations typically follow near-Gaussian behaviour. However, near lock conditions introduce impulsive variations, leading to increased kurtosis values.

The excess kurtosis is the fourth standardized moment of a distribution:

$$\kappa = E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] - 3$$

where μ is the mean and σ is the standard deviation. Kurtosis characterizes the "peakedness" and tail behavior:

- $\kappa < 0$: Platykurtic (flatter than normal distribution)
- $\kappa = 0$: Mesokurtic (normal distribution)
- $\kappa > 0$: Leptokurtic (more peaked, heavier tails)

Dominant Frequency Estimation

Dominant frequency is estimated using zero-crossing techniques to identify oscillatory behaviour in the signal.

Changes in frequency characteristics provide additional indication of abnormal wheel behaviour.

The dominant frequency provides information about the characteristic oscillation timescale:

$$f_{\text{dominant}} = \frac{N_{\text{zero crossings}}}{2T_{\text{window}}} \cdot f_{\text{sample}}$$

where $N_{\text{zero crossings}}$ is the number of times the signal crosses its mean value. This zero-crossing method is computationally efficient (suitable for embedded systems) and captures the main frequency content without requiring FFT computation.

e) Lockup Probability Estimation

A weighted lockup probability is calculated using multiple signal features.

The weighting factors used in this implementation are:

- Entropy contribution: **35%**
- Kurtosis contribution: **25%**
- Frequency contribution: **40%**

These weights are applied to combine feature outputs into a single probability value representing the likelihood of wheel lock.

f) Predictive Warning Generation

Based on computed lockup probability values, predictive warnings are generated before the occurrence of excessive slip.

These warnings are compared against traditional ABS activation signals to evaluate early detection capability.

The predictive method generated **170 predictive events**, compared with conventional reactive triggers in the dataset.

g) Performance Evaluation Module

The final module evaluates predictive ABS performance against traditional ABS behaviour.

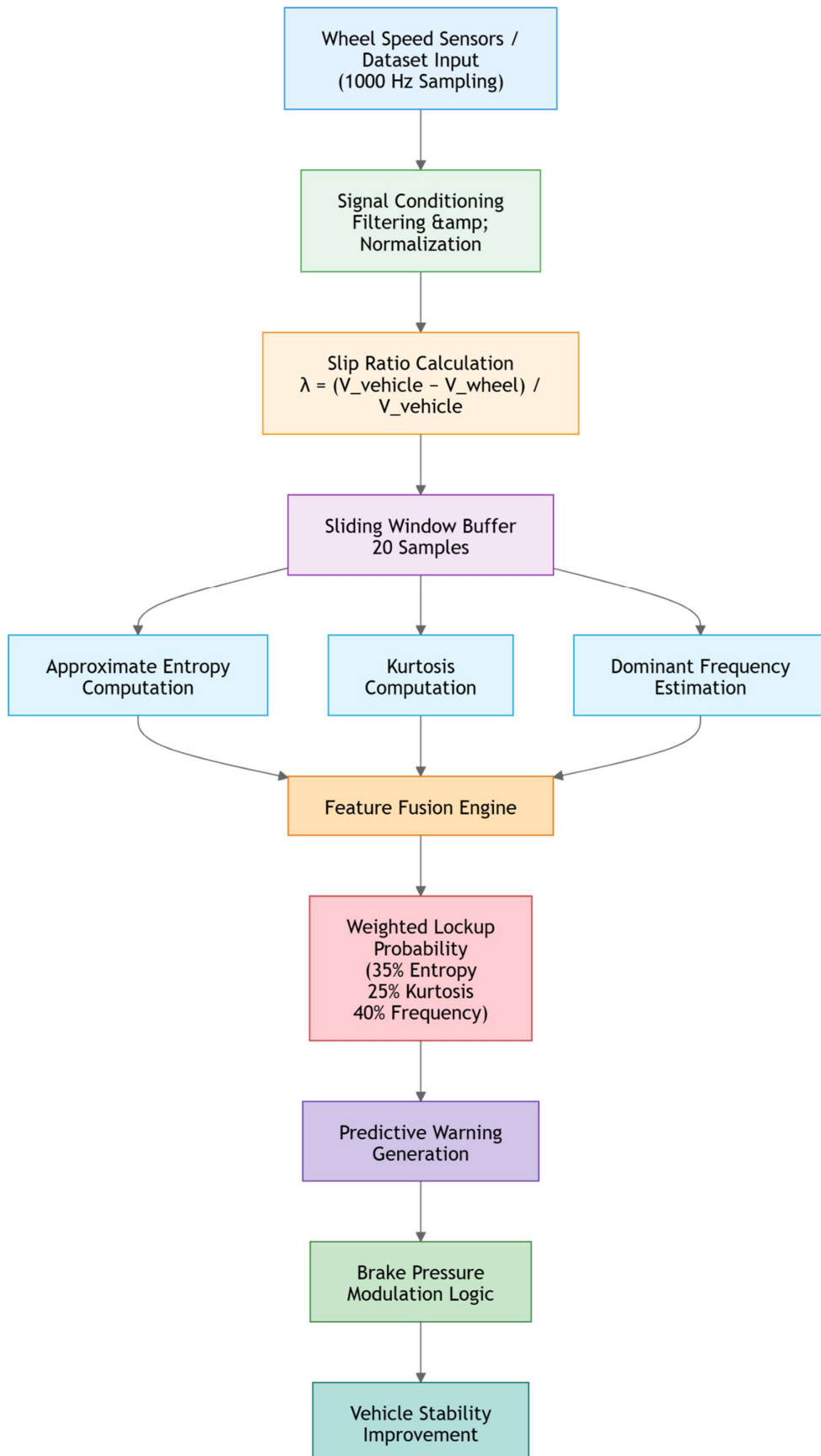
Performance parameters evaluated include:

- Detection latency
- Slip regulation accuracy
- Stopping distance
- Computational overhead

These results are used to assess the effectiveness of predictive feature-based control.

2. FUNCTIONAL BLOCK DIAGRAM / DESCRIPTION

The predictive ABS system operates in a closed-loop configuration that continuously monitors wheel dynamics and generates early warnings when abnormal behaviour is detected.



The generated outputs are compared with conventional ABS activation signals to determine detection latency differences and evaluate improvement in braking response timing.

3. MAJOR TECHNICAL SPECIFICATIONS

SI No	Parameter	Specification
1	Application	Predictive Anti-lock Braking System
2	Data Source	Real braking dataset
3	Sampling Rate	1000 Hz
4	Number of Wheels	Four
5	Buffer Size	20 Samples
6	Feature Extraction	Entropy, Kurtosis, Frequency
7	Lockup Prediction	Weighted multi-feature method
8	Processing Platform	Python-based signal processing
9	Dataset Size	15,000 Samples
10	Slip Range	10% – 20% optimal region
11	Prediction Method	Probability-based detection
12	Output	Predictive warning signal

4. DESIGN CONSIDERATIONS

- Several design considerations were incorporated during system development.
- The system must operate under real-time constraints with limited computational resources. Feature computation methods were therefore selected based on computational efficiency.
- Signal window size was carefully chosen to balance detection accuracy and processing latency.
- Slip ratio limits were maintained within the optimal friction range to prevent excessive tire slip.
- Multiple features were combined to reduce dependency on a single indicator and improve prediction reliability.
- Threshold values were designed to support operation across multiple road surface conditions such as dry, wet, and low-friction surfaces.

5. FUTURE PLAN

Phase	Activity	Status
Literature Review	Study of predictive braking techniques	Completed
Algorithm Development	Implementation of entropy-based detection	In Progress
Dataset Processing	Multi-feature computation	In Progress
Performance Evaluation	Comparative validation	Pending
Real-Time Implementation	Embedded deployment testing	Pending
Final Dissertation	Documentation and submission	Pending